Thiết kế giao diện

User Interface Design Evaluation

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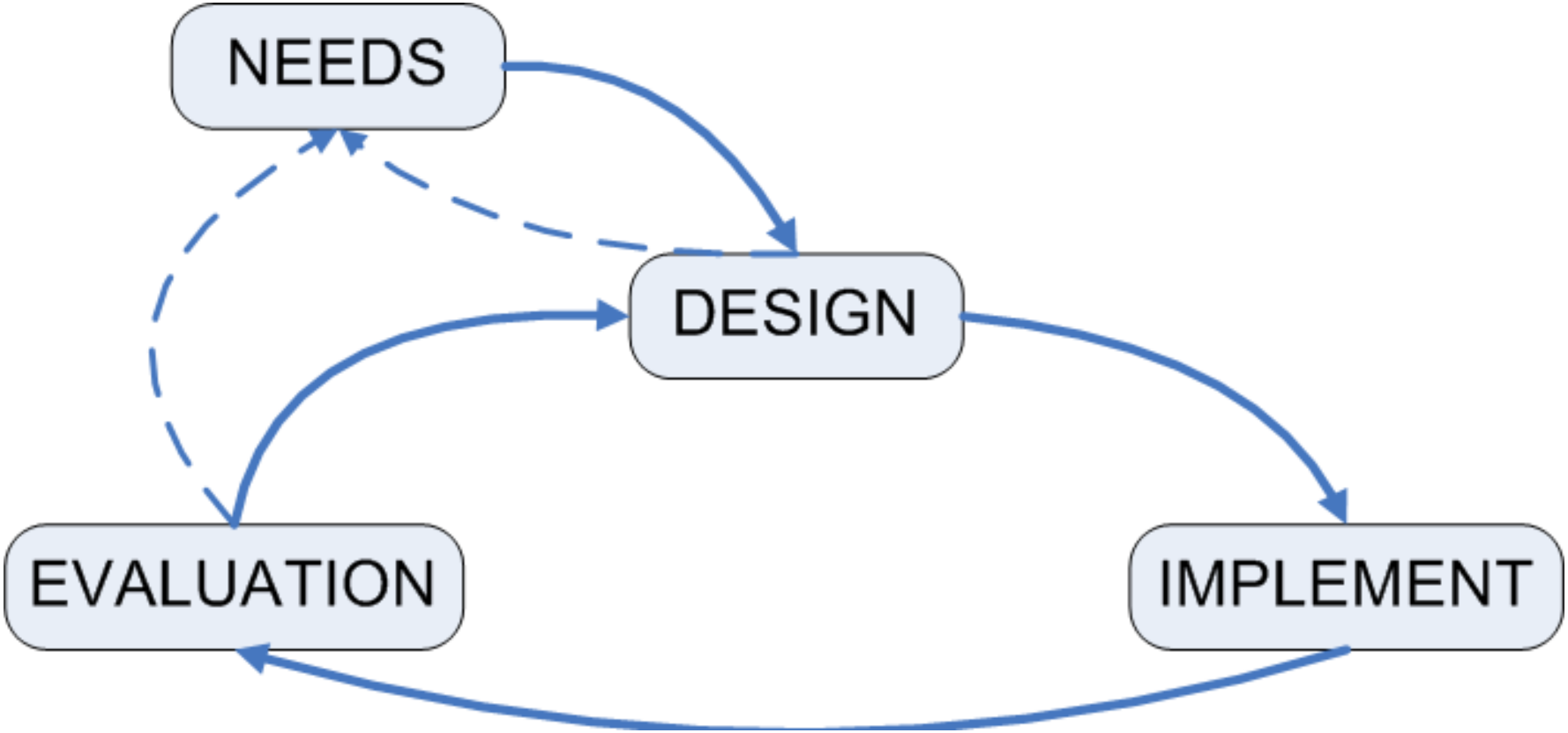
1. UI design
    1. *Design process*
    2. *Scenarios & Use cases*
    3. ***Prototyping***
    4. Evaluation
2. Class project [2]

Content

1. UI design
    1. Design process
    2. Scenarios & Use cases
    3. Prototyping
    4. **Evaluation**
2. Class project [3]

UI Design process



Interaction Design process

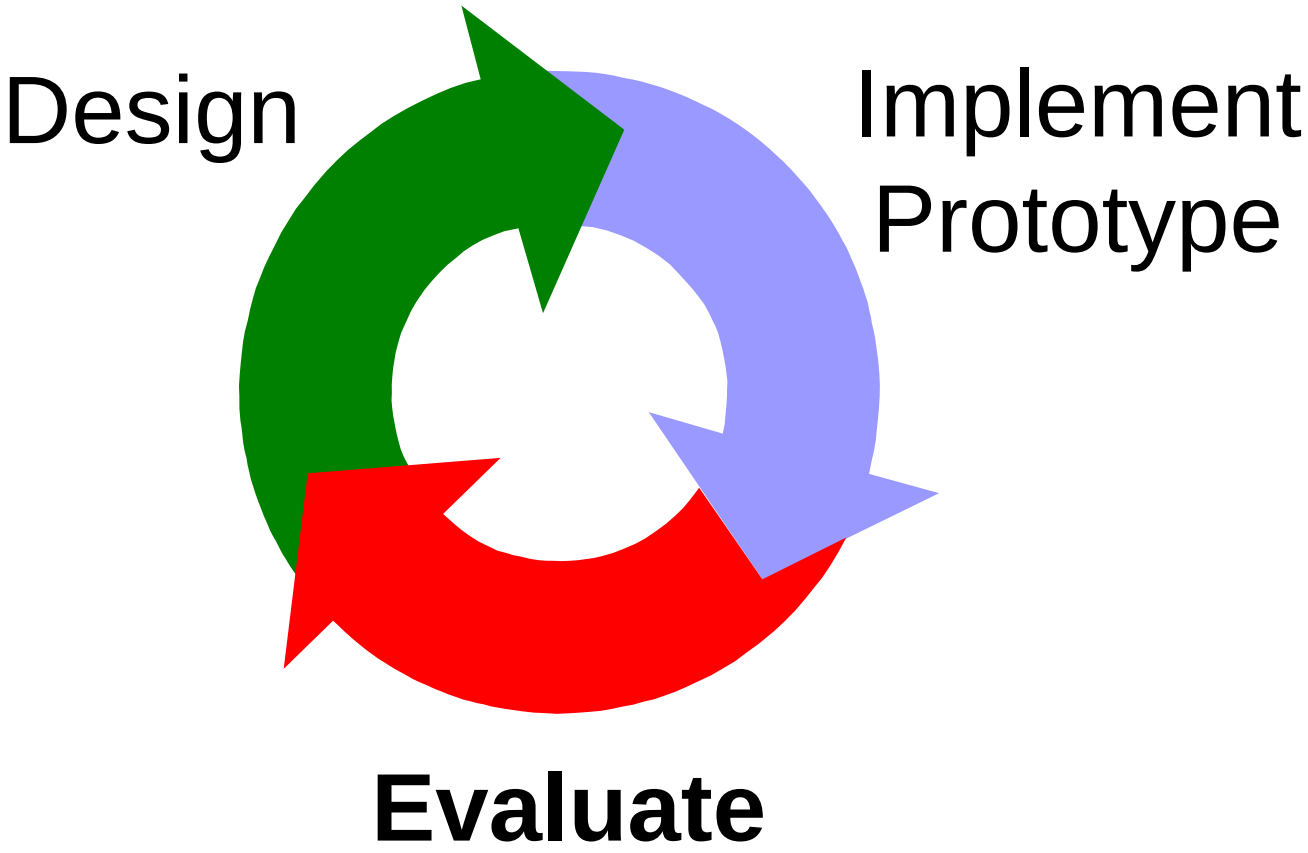
1. Identify needs & establishing requirements.

2. Develop alternative designs that meet those requirements.

3. Building interactive versions of the designs.

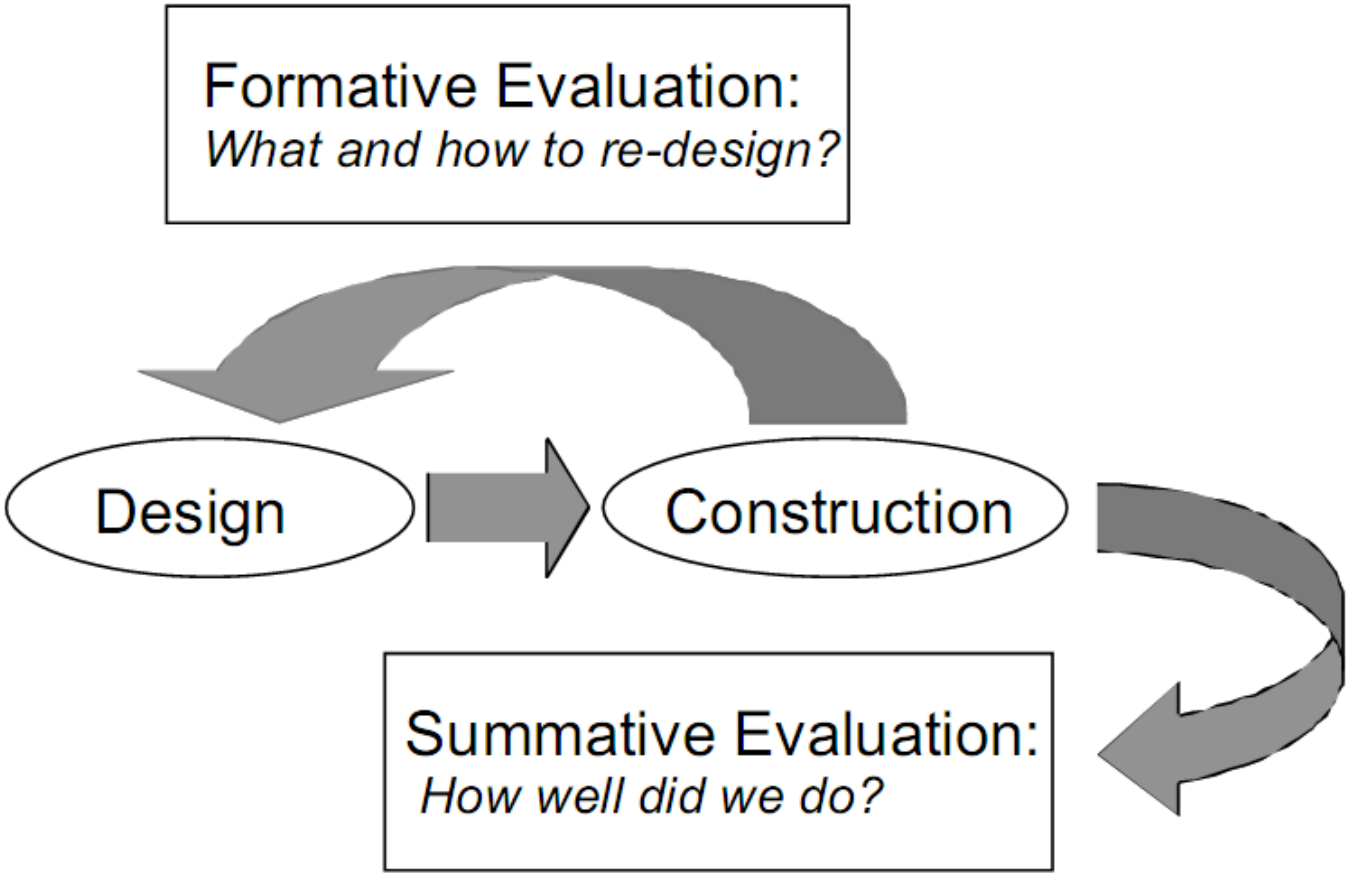
4. **Evaluating what is being built throughout the process.**

UI Design process



Usability evaluation

- Evaluation goals:
  - To provide feedback in design process for better designs
  - To access how usable the product is



Evaluation phases

1. Planning – Training: prepare evaluation

2. Conducting: do evaluation

3. Sumarizing: make evaluation results

4. Debriefing: present & use the results

A framework to guide evaluation: DECIDE

- **D**etermine overall goals
- **E**xplore specific questions to be answered
- **C**hoose the evaluation methods
- **I**dentify the practical issues
- **D**ecide how to deal with ethical issues
- **E**valuate the data

A framework to guide evaluation: DECIDE

Overall goals & Specific questions

Example:

- Goal: why people prefer paper airline tickets to e-tickets
- Question:
  - What do users think about e-tickets?
  - Do they trust the system?
  - Do customers easily access computers to make booking?
  - Are customers think about security?
  - Is the interface so bad?
  - ...

A framework to guide evaluation: DECIDE

Evaluation methods

- Analytic methods
  - Heuristic evaluation
  - Cognitive walkthrough
  - GOM analysis
- Empirical methods
  - Field study
  - Interview
  - Lab-based usability evaluation
  - Controlled experiment

A framework to guide evaluation: DECIDE

Practical issues:

- Evaluators
- Select participants
- Resources: facilities & equipments
- Time & money

Ethical issues:

- Anonymous term
- Rights of users
- Inform consent form

A framework to guide evaluation: DECIDE

Evaluate **data**

- What data to be collected?
- How statistics?
- Bias
- Validity
- Present the results
- Further process…

A framework to guide evaluation: DECIDE

Analyze the **data:**

- Initial reaction:
  - “stupid user!”, “that’s developer X’s fault!”, “this sucks”
- Mature reaction:
  - “how can we redesign UI to solve that usability problem?”
  - the user and/or the data is always ‘right’
- Identify usability problems
  - Learning issues: e.g. can’t figure out or didn’t notice feature
  - Performance issues: e.g. arduous, tiring to solve tasks
  - Subjective issues: e.g. annoying, ugly
- Problem severity: critical vs. minor
- Redesign

A framework to guide evaluation: DECIDE

Analyze the **data:** Problem severity: critical vs. minor

Problem	Importance	Solutions	Cost	Ratio I/C

- Importance 1-5: (task effect, frequency)
  - 5 = critical, major impact on user, frequent occurrence
  - 3 = user can complete task, but with difficulty
  - 1 = minor problem, small speed bump, infrequent
- Ratio = importance / cost
  - Sort by this, highest to lowest
  - 3 categories: Must fix, next version, ignored

A framework to guide evaluation: DECIDE

Analyze the **data:** Examples

E.g.1. Problem: user didn't know he could zoom in

Potential solutions:

+ Better labeling: Better zoom button icon, tooltip
+ Clearer affordance: Add a zoom bar slider (like Google maps)
+ …
+ NOT: more "help" documentation! You can do better.

E.g.2. Problem: user took too long to repeatedly zoom in

Potential solutions:

+ Faster affordance: Add a real-time zoom bar
+ Shortcuts: Icons for each zoom level: state, city, street
+ …

Evaluation methods

About evaluation methods:

- Analytic methods
  - Heuristic evaluation
  - Cognitive walkthrough
  - GOM analysis
- Empirical methods
  - Query
  - Field study
  - Lab-based usability evaluation

Evaluation methods

- Analytic methods:
  - Guideline – based evaluation: rapid, cheap but may miss realistic details.
  - Model-base: good scientific foundation, can be powerful but take time, money and require expertise
- Empirical methods:
  - Conclusions based on actual use but the participants might not representative, the test environment might not large, diverse enough…

Evaluation method – Heuristic evaluation

Evaluate based on Jacob Nielsen's 10 Usability Heuristics

1. Visibility of system status.
2. Match between system and real world.
3. User control and freedom.
4. Consistency and standards.
5. Error Prevention.
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Help users recognize, diagnose, and recover from errors
10. Help and Documentation

- Do not require users
- 3-5 evaluators work independently on UI

Evaluation method – Heuristic evaluation

Heuristic	Severity	Description
Consistency	3	The interface used the string "Save" on the first screen for saving the user's file, but used the string "Write file" on the second screen. Users may be confused by this different terminology for the same function
Error Message	4	Entering invalid input into dialog box on first form results in “Error 103”

No.	*Heuristic*	*Location*	*Description*	*Severity*	*Fixability*	*Sum*	*Possible Fix*
1	Visibility of system status	Home screen	The home screen does not portray any information about battery power remaining, making it hard for users to tell how much power they have left.	3	3	6	Display a battery life indicator on the home screen.
2	User control and freedom	Screen for writing a text message	Once you are on the screen for writing a text message, you cannot leave without sending the message. Users need a way to get out if they decide not to send a message.	3	2	5	Allow the CLR button to always move the user back one screen no matter where they are.

Evaluation methods

- *Cognitive walkthrough*
- *Grade of membership (GOM) analysis*

→ *Predictive evaluation*

Self - study

Evaluation method – Query user

- Questionaires and/or Interview
- Prepared questions:
  - Clear & simple questions
  - Avoid complex questions
  - The ordering is important
  - When using scales, be careful on range
  - Avoid jargon
  - Avoid subjective questions
- Provide clear instruction
- Online questionaires (e.g. http://www.surveymonkey.com/)

Evaluation method – Question Examples

- TaiLieu\user_inspection_checklist.pdf
- Scale question:

How easy was the system to use?

Easy — Difficult

1 2 3 4 5 6 7

The system was easy to use

Strongly Agree | Agree | Neutral | Disagree | Strongly Disagree

How did you feel about the ease of using the system?

Evaluation method – Question Examples

- Bad questions:
  - This is a ugly website, isn’t it?
  - If you answered yes to the last question then please go to question 74, unless you gave a score of less than 5 for questions 5-8, in which case go to question 78.
  - Is it easier to navigate if the menu is on left-side?

Evaluation method – Question Examples

- Bad question design:
  - Exact age
  - Overlap range
  - Too small space to write answer.

2. State your age in years ☐

3. How long have you used the Internet? *(check one only)*
   - ☐ <1 year
   - ☐ 1–3 years
   - ☐ 3–5 years
   - ☐ >5 years

4. Do you use the Web to:
   - purchase goods ☐
   - send e-mail ☐
   - visit chatrooms ☐
   - use bulletin boards ☐
   - find information ☐
   - read the news ☐

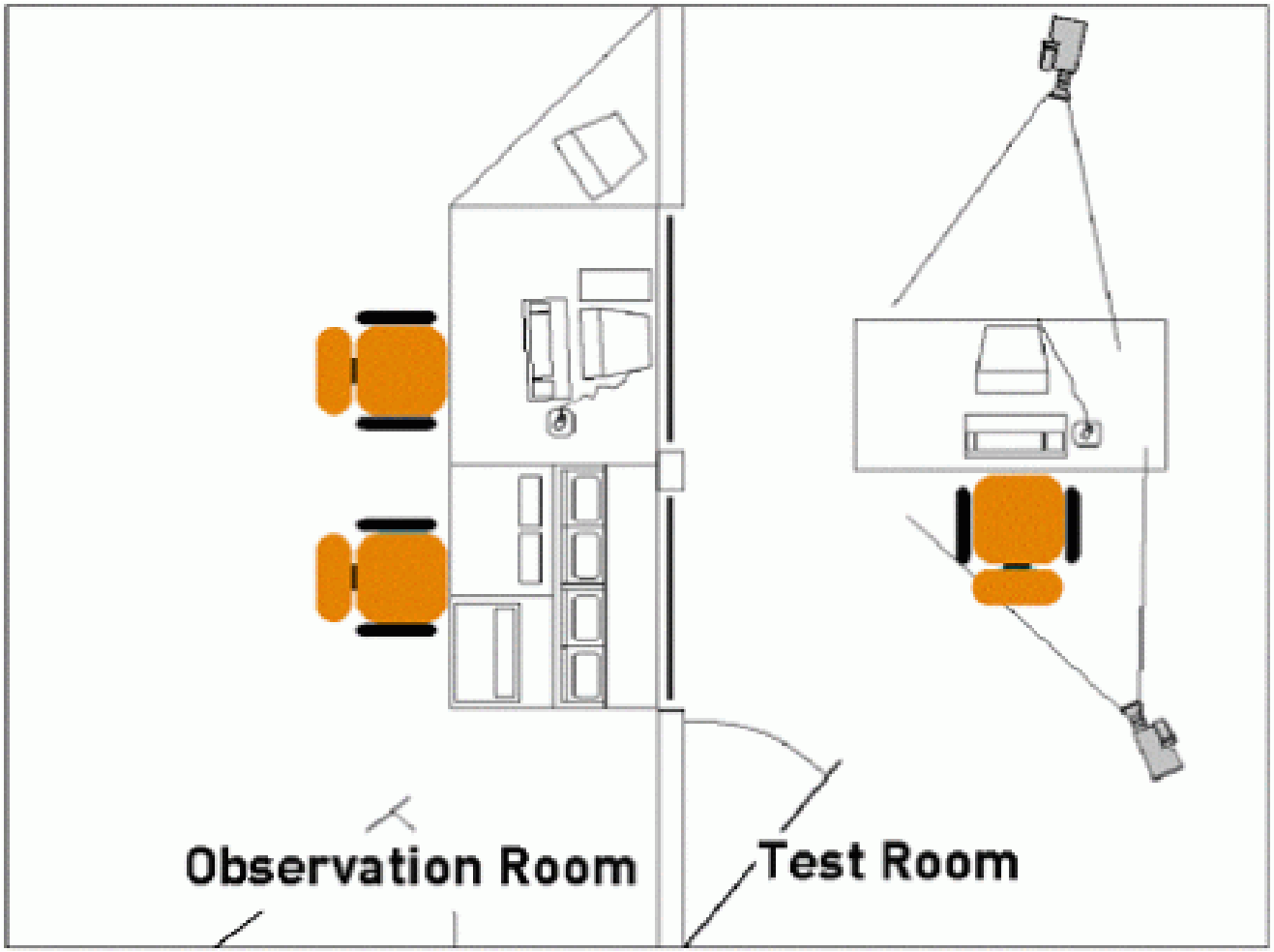
5. How useful is the Internet to you?

_____

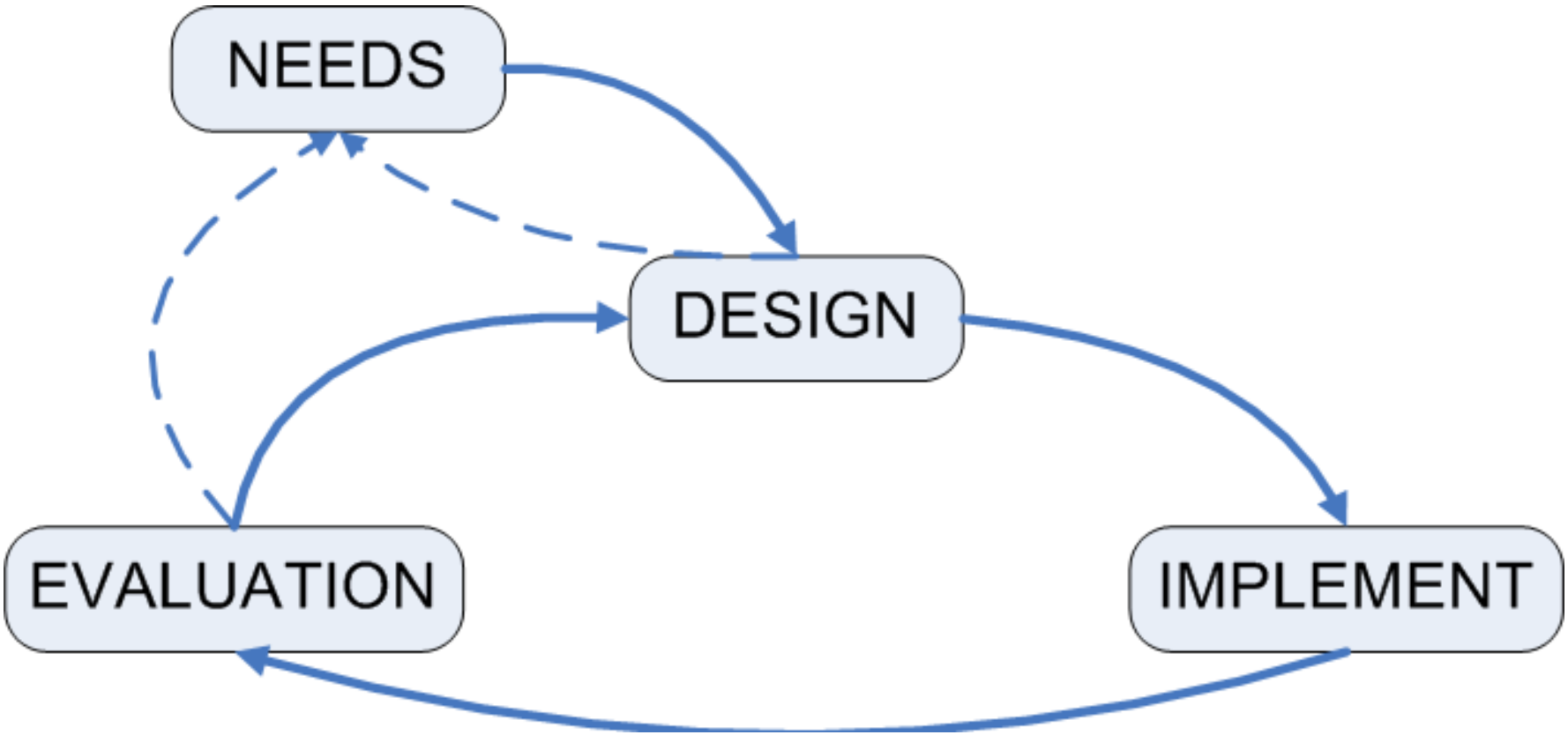
_____

Evaluation method – Lab-based evaluation

Lab-based usability evaluation



UI Design process



Interaction Design process

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Prototype

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Evaluation result

References

- http://www.sciencebuddies.org/science-fair-projects/project_ideas/Soc_survey.shtml

Thực hành

- Bài tập [3] : khảo sát đánh giá và cải tiến giao diện webpage

  BaiTap\BaiTap_3.doc

  Có thể chọn website để cải tiến từ http://www.webpagesthatsuck.com/ mục Worst web sites

Thực hành - Bài tập đánh giá, cải tiến giao diện website

Gợi ý: website http://www.findapetonline.com/

Chức năng: website giới thiệu sản phẩm vật nuôi

Thực hành - Bài tập đánh giá, cải tiến giao diện website

Gợi ý: website http://www.findapetonline.com/

- ~50% câu hỏi về thiết kế chung và chức năng chung của website

VD: + Bạn thấy màu sắc (màu chữ, màu nền…) của website thế nào?

+ Đánh giá của bạn về font chữ menu của website:

1 2 3 4 5

(Rất dễ nhìn) (Rất khó nhìn)

+ Phong cách canh giữa chữ của website làm cho bạn:

1 2 3 4 5

(Rất dễ đọc) (Rất khó đọc)

+ Liệt kê các sản phẩm, dịch vụ mà website cung cấp. Vui lòng cung cấp thời gian để bạn tìm được các sản phẩm, dịch vụ đã liệt kê.

Thực hành - Bài tập đánh giá, cải tiến giao diện website

Gợi ý: website http://www.findapetonline.com/

- ~50% câu hỏi về việc **sử dụng** website: Xây dựng những kịch bản (scenario) sử dụng website, yêu cầu user làm theo kịch bản và xây dựng câu hỏi dựa trên những kịch bản này.

VD: Scenerio: Một user truy cập website, xem thông tin về chó con đang được bán. Sau đó xem thông tin về mèo con và chọn được một con mèo muốn mua. User này tìm số điện thoại trên website để liên lạc, hỏi giá con mèo muốn mua.

+ Bạn hãy vào trang giới thiệu chó con và xem thông tin của các chó con được giới thiệu. Bạn muốn thêm thông tin gì?

Thực hành - Bài tập đánh giá, cải tiến giao diện website

+ Hãy nhận xét về cách trình bày thông tin chó con:

1	2	3	4	5
(Rất tốt)				(Rất kém)

+ Bạn hãy vào trang giới thiệu chó con và tìm xem trang này có được cập nhật thông tin không? Tại sao bạn có câu trả lời?

+ Bạn hãy vào trang giới thiệu mèo con. Giả sử bạn thích một con mèo, hãy tìm số điện thoại để liên lạc với cửa hàng; cho biết cách bạn tìm được số điện thoại này.

+ Giả sử bạn thích một con chó, hoặc mèo trên website. Khi nói chuyện điện thoại với nhân viên cửa hàng, bạn sẽ xác định con vật bạn muốn hỏi thông tin như thế nào? (Vd: gọi tên, đếm STT)